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Image sensors having high dynamic range pixels

Abstract

An image sensor may include an array of imaging pixels arranged in rows and columns. Each imaging pixel may include a photodiode, an overflow capacitor, an overflow transistor that is interposed between the photodiode and the overflow capacitor, a floating diffusion region, a transfer transistor that is interposed between the photodiode and the floating diffusion region, a voltage supply, and a reset transistor that is interposed between the floating diffusion region and the voltage supply. The voltage supply may provide a voltage at a first magnitude that is less than the pinning voltage for a first portion of a reset period and may provide the voltage at a second magnitude that is greater than the pinning voltage for a second portion of the reset period.

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Background/Summary

BACKGROUND

(1) This relates generally to imaging devices, and more particularly, to imaging devices having high dynamic range imaging pixels.

(2) Image sensors are commonly used in electronic devices such as cellular telephones, cameras, and computers to capture images. In a typical arrangement, an image sensor includes an array of image pixels arranged in pixel rows and pixel columns. Circuitry may be coupled to each pixel

column for reading out image signals from the image pixels.
(3) Typical image pixels contain a photodiode for generating charge in response to incident light. Image sensors can operate using a global shutter or a rolling shutter scheme. In a global shutter, every pixel in the image sensor may simultaneously capture an image, whereas in a rolling shutter each row of pixels may sequentially capture an image.
(4) Some conventional image sensors may be able to operate in a high dynamic range (HDR) mode. HDR operation may be accomplished in image sensors by assigning alternate rows of pixels different integration times. However, conventional image sensors may sometimes experience lower than desired resolution, lower than desired sensitivity, higher than desired noise levels, and lower than desired quantum efficiency.
(5) It is within this context that the embodiments described herein arise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram of an illustrative electronic device having an image sensor in accordance with some embodiments.
(2) FIG. **2** is a diagram of an illustrative pixel array and associated row and column control circuitry for reading out image signals from an image sensor in accordance with some embodiments.
(3) FIG. **3** is a circuit diagram of an illustrative imaging pixel that includes an overflow capacitor in accordance with some embodiments.
(4) FIG. **4** is a circuit diagram of an illustrative imaging pixel that includes multiple sub-pixels with respective overflow capacitors in accordance with some embodiments.
(5) FIG. **5** is a timing diagram showing an illustrative method of operation for an imaging pixel such as the imaging pixel in FIG. **4** in accordance with some embodiments.
(6) FIG. **6** is a circuit diagram of an illustrative imaging pixel that includes an overflow capacitor and a dual conversion gain capacitor in accordance with some embodiments.
(7) FIG. **7** is a circuit diagram of an illustrative imaging pixel that includes multiple sub-pixels with respective overflow capacitors and a shared dual conversion gain capacitor in accordance with some embodiments.
(8) FIG. **8** is a circuit diagram of an illustrative imaging pixel that includes overflow capacitors coupled to a photodiode in parallel in accordance with some embodiments.
(9) FIG. **9** is a circuit diagram of an illustrative imaging pixel that includes overflow capacitors coupled to a photodiode in series in accordance with some embodiments.
(10) FIG. **10** is a circuit diagram of an illustrative imaging pixel that includes a coupled gate structure in accordance with some embodiments.
(11) FIG. **11** is a circuit diagram of an illustrative imaging pixel that includes an overflow capacitor and a calibration transistor in accordance with some embodiments.
(12) FIG. **12** is a timing diagram showing an illustrative method of operation for an imaging pixel such as the imaging pixel in FIG. **11** in accordance with some embodiments.
DETAILED DESCRIPTION
(13) Embodiments of the present technology relate to image sensors. It will be recognized by one skilled in the art that the present exemplary embodiments may be practiced without some or all of these specific details. In other instances, well-known operations have not been described in detail in order not to unnecessarily obscure the present embodiments.
(14) Electronic devices such as digital cameras, computers, cellular telephones, and other electronic devices may include image sensors that gather incoming light to capture an image. The image sensors may include arrays of pixels. The pixels in the image sensors may include photosensitive

elements such as photodiodes that convert the incoming light into image signals. Image sensors may have any number of pixels (e.g., hundreds or thousands or more). A typical image sensor may, for example, have hundreds or thousands or millions of pixels (e.g., megapixels). Image sensors may include control circuitry such as circuitry for operating the pixels and readout circuitry for reading out image signals corresponding to the electric charge generated by the photosensitive elements.

(15) FIG. **1** is a diagram of an illustrative imaging and response system including an imaging system that uses an image sensor to capture images. System **100** of FIG. **1** may be an electronic device such as a camera, a cellular telephone, a video camera, or other electronic device that captures digital image data, may be a vehicle safety system (e.g., an active braking system or other vehicle safety system), or may be a surveillance system.

(16) As shown in FIG. **1**, system **100** may include an imaging system such as imaging system **10** and host subsystems such as host subsystem **20**. Imaging system **10** may include camera module **12**. Camera module **12** may include one or more image sensors **14**, such as in an image sensor array integrated circuit, and one or more lenses.

(17) During image capture operations, each lens may focus light onto an associated image sensor **14**. Image sensor **14** may include photosensitive elements (i.e., image sensor pixels) that convert the light into digital data. Image sensors may have any number of pixels (e.g., hundreds, thousands, millions, or more). A typical image sensor may, for example, have millions of pixels (e.g., megapixels).

(18) Each image sensor in camera module **12** may be identical or there may be different types of image sensors in a given image sensor array integrated circuit. In some examples, image sensor **14** may further include bias circuitry (e.g., source follower load circuits), sample and hold circuitry, correlated double sampling (CDS) circuitry, amplifier circuitry, analog-to-digital converter circuitry, data output circuitry, memory (e.g., buffer circuitry), and/or address circuitry.

(19) Still and video image data from sensor **14** may be provided to image processing and data formatting circuitry **16** via path **28**. Image processing and data formatting circuitry **16** may be used to perform image processing functions such as data formatting, adjusting white balance and exposure, implementing video image stabilization, or face detection. Image processing and data formatting circuitry **16** may additionally or alternatively be used to compress raw camera image files if desired (e.g., to Joint Photographic Experts Group or JPEG format).

(20) In one example arrangement, such as a system on chip (SoC) arrangement, sensor **14** and image processing and data formatting circuitry **16** are implemented on a common semiconductor substrate (e.g., a common silicon image sensor integrated circuit die). If desired, sensor **14** and image processing circuitry **16** may be formed on separate semiconductor substrates. For example, sensor **14** and image processing circuitry **16** may be formed on separate substrates that have been stacked.

(21) Imaging system **10** may convey acquired image data to host subsystem **20** over path **18**. Host subsystem **20** may include input-output devices **22** and storage processing circuitry **24**. Host subsystem **20** may include processing software for detecting objects in images, detecting motion of objects between image frames, determining distances to objects in images, or filtering or otherwise processing images provided by imaging system **10**. For example, image processing and data formatting circuitry **16** of the imaging system **10** may communicate the acquired image data to storage and processing circuitry **24** of the host subsystems **20**.

(22) If desired, system **100** may provide a user with numerous high-level functions. In a computer or cellular telephone, for example, a user may be provided with the ability to run user applications. For these functions, input-output devices **22** of host subsystem **20** may include keypads, input-output ports, buttons, and displays and storage and processing circuitry **24**. Storage and processing circuitry **24** of host subsystem **20** may include volatile and/or nonvolatile memory (e.g., random-access memory, flash memory, hard drives, solid-state drives, etc.). Storage and processing

circuitry **24** may additionally or alternatively include microprocessors, microcontrollers, digital signal processors, and/or application specific integrated circuits.
(23) An example of an arrangement of image sensor **14** of FIG. **1** is shown in FIG. **2**. As shown in FIG. **2**, image sensor **14** may include control and processing circuitry **44**. Control and processing circuitry **44** (sometimes referred to as control and processing logic) may be part of image processing and data formatting circuitry **16** in FIG. **1** or may be separate from circuitry **16**. Image sensor **14** may include a pixel array such as array **32** of pixels **34** (sometimes referred to herein as image sensor pixels, imaging pixels, or image pixels). Control and processing circuitry **44** may be coupled to row control circuitry **40** via control path **27** and may be coupled to column control and readout circuits **42** via data path **26**.
(24) Row control circuitry **40** may receive row addresses from control and processing circuitry **44** and may supply corresponding row control signals to image pixels **34** over one or more control paths **36**. The row control signals may include pixel reset control signals, charge transfer control signals, blooming control signals, row select control signals, dual conversion gain control signals, or any other desired pixel control signals.
(25) Column control and readout circuitry **42** may be coupled to one or more of the columns of pixel array **32** via one or more conductive lines such as column lines **38**. A given column line **38** may be coupled to a column of image pixels **34** in image pixel array **32** and may be used for reading out image signals from image pixels **34** and for supplying bias signals (e.g., bias currents or bias voltages) to image pixels **34**. In some examples, each column of pixels may be coupled to a corresponding column line **38**. For image pixel readout operations, a pixel row in image pixel array **32** may be selected using row driver circuitry **40** and image data associated with image pixels **34** of that pixel row may be read out by column readout circuitry **42** on column lines **38**. Column readout circuitry **42** may include column circuitry such as column amplifiers for amplifying signals read out from array **32**, sample and hold circuitry for sampling and storing signals read out from array **32**, analog-to-digital converter circuits for converting read out analog signals to corresponding digital signals, or column memory for storing the readout signals and any other desired data. Column control and readout circuitry **42** may output digital pixel readout values to control and processing logic **44** over line **26**.
(26) Array **32** may have any number of rows and columns. In general, the size of array **32** and the number of rows and columns in array **32** will depend on the particular implementation of image sensor **14**. While rows and columns are generally described herein as being horizontal and vertical, respectively, rows and columns may refer to any grid-like structure. Features described herein as rows may be arranged vertically and features described herein as columns may be arranged horizontally.
(27) Pixel array **32** may be provided with a color filter array having multiple color filter elements which allows a single image sensor to sample light of different colors. As an example, image sensor pixels such as the image pixels in array **32** may be provided with a color filter array which allows a single image sensor to sample red, green, and blue (RGB) light using corresponding red, green, and blue image sensor pixels. The red, green, and blue image sensor pixels may be arranged in a Bayer mosaic pattern. The Bayer mosaic pattern consists of a repeating unit cell of two-by-two image pixels, with two green image pixels diagonally opposite one another and adjacent to a red image pixel diagonally opposite to a blue image pixel. In another example, broadband image pixels having broadband color filter elements (e.g., clear color filter elements, yellow color filter elements, etc.) may be used instead of green pixels in a Bayer pattern. These examples are merely illustrative and, in general, color filter elements of any desired color and in any desired pattern may be formed over any desired number of image pixels **34**.
(28) In some implementations, array **32** may be part of a stacked-die arrangement in which pixels **34** of array **32** are split between two or more stacked substrates. In such an arrangement, each of the pixels **34** in the array **32** may be split between the two dies at any desired node within the pixel.

As an example, a node such as the floating diffusion node may be formed across two dies. Pixel circuitry that includes the photodiode and the circuitry between the photodiode and the desired node (such as the floating diffusion node, in the present example) may be formed on a first die, and the remaining pixel circuitry may be formed on a second die. The desired node may be formed on (i.e., as a part of) a coupling structure (such as a conductive pad, a micro-pad, a conductive interconnect structure, or a conductive via) that connects the two dies. Before the two dies are bonded, the coupling structure may have a first portion on the first die and may have a second portion on the second die. The first die and the second die may be bonded to each other such that first portion of the coupling structure and the second portion of the coupling structure are bonded together and are electrically coupled. If desired, the first and second portions of the coupling structure may be compression bonded to each other. However, this is merely illustrative. If desired, the first and second portions of the coupling structures formed on the respective first and second dies may be bonded together using any metal-to-metal bonding technique, such as soldering or welding.

(29) As mentioned above, the desired node in the pixel circuit that is split across the two dies may be a floating diffusion node. Alternatively, the desired node in the pixel circuit that is split across the two dies may be any other node along the pixel circuit. In one alternative, the desired node split across two dies may be the node between a floating diffusion region and the gate of a source follower transistor. For example, the floating diffusion node may be formed on the first die on which the photodiode is formed, while the coupling structure may connect the floating diffusion node to the source follower transistor on the second die. In another alternative, the desired node split across two dies may be the node between a floating diffusion region and a source-drain node of a transfer transistor. For example, the floating diffusion node may be formed on the second die on which the photodiode is not located. In yet another alternative, the desired node split across two dies may be the node between a source-drain node of a source follower transistor and a row select transistor.

(30) In general, array **32**, row control circuitry **40**, and column control and readout circuitry **42** may be split between two or more stacked substrates. In one example, array **32** may be formed in a first substrate and row control circuitry **40** and column control and readout circuitry **42** may be formed in a second substrate. In another example, array **32** may be split between first and second substrates (using one of the pixel splitting schemes described above) and row control circuitry **40** and column control and readout circuitry **42** may be formed in a third substrate. In other examples, row control circuitry **40** may be on a separate substrate from column control and readout circuitry **42**. In yet another example, row control circuitry **40** may be split between two or more substrates and/or column control and readout circuitry **42** may be split between two or more substrates.

(31) FIG. **3** is a circuit diagram of an imaging pixel having a photosensitive element and a storage capacitor. As shown in FIG. **3**, image pixel **34** includes photosensitive element **102** (e.g., a photodiode) that generates charge in response to incident light. Photosensitive element **102** has a first terminal that is coupled to ground. Photosensitive element **102** may be a pinned photodiode with a corresponding pinning voltage. The second terminal of photosensitive element **102** is coupled to transfer transistor **104** and transistor **105**. Transfer transistor **104** is coupled to floating diffusion (FD) region **118**. Transistor **105** (sometimes referred to as threshold transistor **105** or overflow transistor **105**) is coupled between photodiode **102** and storage capacitor **110** (sometimes referred to as overflow capacitor **110**). Capacitor **110** has a first plate coupled to transistor **105** and a second plate coupled to a voltage supply **122**. Herein, a voltage supply may sometimes alternatively be referred to as a voltage supply terminal, bias voltage supply terminal, bias voltage supply, etc. A reset transistor **106** may be coupled between floating diffusion region **118** and voltage supply **124-2**. Voltage supply **122** may provide an adjustable voltage CLG_REF whereas voltage supply **124-2** may provide an adjustable voltage RST_D. Floating diffusion region **118** may be a doped semiconductor region (e.g., a region in a silicon substrate that is doped by ion

implantation, impurity diffusion, or other doping process). Floating diffusion **118** has an associated capacitance.

(32) Source follower transistor **112** (SF) has a gate terminal coupled to floating diffusion region **118** and a first terminal of reset transistor **106**. Source follower transistor **112** also has a first source-drain terminal coupled to voltage supply **124-3**. In this application, each transistor is illustrated as having three terminals: a source, a drain, and a gate. The source and drain terminals of each transistor may be changed depending on how the transistors are biased and the type of transistor used. For the sake of simplicity, the source and drain terminals are referred to herein as source-drain terminals or simply terminals. Voltage supply **124-3** may provide a power supply voltage VDD. A second source-drain terminal of source follower transistor **112** is coupled to transistor **132**.

(33) Transistor **132** (sometimes referred to as a row select transistor) is interposed between source follower transistor **112** and column output line **116**. An output voltage (PIXOUT) is provided on column output line **116** when row select transistor **132** is asserted.

(34) Source follower transistor **112**, row select transistor **132**, and column output line **116** may sometimes collectively be referred to as a readout circuit or as readout circuitry. Other readout circuits may be used if desired.

(35) A gate terminal of overflow transistor **105** receives control signal OG. A gate terminal of transfer transistor **104** receives control signal TX. A gate terminal of reset transistor **106** receives control signal RST. A gate terminal of row select transistor **132** receives control signal RS. Control signals OG, TX, RST, and RS may be provided by row control circuitry (e.g., row control circuitry **40** in FIG. **2**) over control paths (e.g., control paths **36** in FIG. **2**). The adjustable voltages at power supplies **122** and **124-2** may also be provided by row control circuitry if desired.

(36) The pixel of FIG. **3** has one photosensitive portion (with photodiode **102**) for a corresponding readout circuit. This example is merely illustrative. If desired, multiple photosensitive portions having the same arrangement as in FIG. **3** may share a single readout circuit. FIG. **4** is a circuit diagram of an imaging pixel having four photosensitive portions (each including a respective photosensitive element and a respective storage capacitor). The four photosensitive portions in the imaging pixel of FIG. **4** all share a common readout circuit. The photosensitive portions are sometimes referred to herein as sub-pixels. The example of four sub-pixels sharing a common readout circuit in FIG. **4** is merely illustrative. In general, any desired number of sub-pixels in any desired layout may share a common readout circuit.

(37) As shown in FIG. **4**, a first photosensitive portion of imaging pixel **34** (sometimes referred to as a first sub-pixel **52-0**) includes a photodiode **102-0**, transfer transistor **104-0**, overflow transistor **105-0**, and storage capacitor **110-0**. The arrangement of these components is the same in FIG. **4** as in FIG. **3** and will therefore not be described again.

(38) A gate terminal of overflow transistor **105-0** receives control signal OG-**0**. A gate terminal of transfer transistor **104-0** receives control signal TX-**0**. Capacitor **110-0** is coupled between transistor **105-0** and a voltage supply **122-0** that provides a voltage CLG_REF_**0**. Control signals OG-**0** and TX-**0** may be provided by row control circuitry (e.g., row control circuitry **40** in FIG. **2**) over control paths (e.g., control paths **36** in FIG. **2**).

(39) A second photosensitive portion of imaging pixel **34** (sometimes referred to as a second sub-pixel **52-1**) includes a photodiode **102-1**, transfer transistor **104-1**, overflow transistor **105-1**, and storage capacitor **110-1**. The arrangement of these components is the same in FIG. **4** as in FIG. **3** and will therefore not be described again.

(40) A gate terminal of overflow transistor **105-1** receives control signal OG-**1**. A gate terminal of transfer transistor **104-1** receives control signal TX-**1**. Capacitor **110-1** is coupled between transistor **105-1** and a voltage supply **122-1** that provides a voltage CLG_REF_**1**. Control signals OG-**1** and TX-**1** may be provided by row control circuitry (e.g., row control circuitry **40** in FIG. **2**) over control paths (e.g., control paths **36** in FIG. **2**).

(41) A third photosensitive portion of imaging pixel **34** (sometimes referred to as a third sub-pixel **52-2**) includes a photodiode **102-2**, transfer transistor **104-2**, overflow transistor **105-2**, and storage capacitor **110-2**. The arrangement of these components is the same in FIG. **4** as in FIG. **3** and will therefore not be described again.
(42) A gate terminal of overflow transistor **105-2** receives control signal OG-**2**. A gate terminal of transfer transistor **104-2** receives control signal TX-**2**. Capacitor **110-2** is coupled between transistor **105-2** and a voltage supply **122-2** that provides a voltage CLG_REF_**2**. Control signals OG-**2** and TX-**2** may be provided by row control circuitry (e.g., row control circuitry **40** in FIG. **2**) over control paths (e.g., control paths **36** in FIG. **2**).
(43) A fourth photosensitive portion of imaging pixel **34** (sometimes referred to as a fourth sub-pixel **52-3**) includes a photodiode **102-3**, transfer transistor **104-3**, overflow transistor **105-3**, and storage capacitor **110-3**. The arrangement of these components is the same in FIG. **4** as in FIG. **3** and will therefore not be described again.
(44) A gate terminal of overflow transistor **105-3** receives control signal OG-**3**. A gate terminal of transfer transistor **104-3** receives control signal TX-**3**. Capacitor **110-3** is coupled between transistor **105-3** and a voltage supply **122-3** that provides a voltage CLG_REF_**3**. Control signals OG-**3** and TX-**3** may be provided by row control circuitry (e.g., row control circuitry **40** in FIG. **2**) over control paths (e.g., control paths **36** in FIG. **2**).
(45) Each one of the transfer transistors in FIG. **4** is coupled to a common floating diffusion region **118**. A reset transistor is coupled between the floating diffusion region and voltage supply **124-2**. A source follower transistor **112** is coupled to floating diffusion region **118**. The arrangement of floating diffusion region **118**, reset transistor **106**, voltage supplies **124-2** and **124-3**, row select transistor **132**, and column output line **116** is the same in FIG. **4** as in FIG. **3** and will not be described again.
(46) Including multiple sub-pixels with a shared readout circuit (as in FIG. **4**) may allow for each sub-pixel to have a small size. The small size may allow for a high resolution of photosensitive regions in the image sensor.
(47) FIG. **5** is a timing diagram showing illustrative operation of the pixel in FIG. **4**. First, there is a reset period up to and including t.sub.3. During the reset period, at t.sub.0, voltage RST_D (from voltage supply **124-2**) is dropped low. Simultaneously, at t.sub.0, the control signal RST is high (meaning that transistor **106** is asserted), the control signals TX_**0**, TX_**1**, TX_**2**, and TX_**3** are high (meaning that transfer transistors **104-0**, **104-1**, **104-2**, and **104-3** are asserted), the control signals OG_**0**, OG_**1**, OG_**2**, and OG_**3** are high (meaning that overflow transistors **105-0**, **105-1**, **105-2**, and **105-3** are asserted), and voltages CLG_REF_**0**, CLG_REF_**1**, CLG_REF_**2**, and CLG_REF_**3** (at voltage supplies **122-0**, **122-1**, **122-2**, and **122-3**, respectively) are high.
(48) For simplicity, the subsequent operations will now be described in detail with respect to a single sub-pixel (e.g., sub-pixel **52-0** with photodiode **102-0**, capacitor **110-0**, transistor **104-0**, and transistor **105-0** in FIG. **4**). However, it should be understood that, during the reset period and integration period, the same operations are simultaneously or consecutively being performed for the other sub-pixels.
(49) Between t.sub.0 and t.sub.1, capacitor **110-0** is electrically connected to voltage supply **124-2** through transistor **105-0**, photodiode **102-0**, transistor **104-0**, floating diffusion region **118**, and reset transistor **106**. When voltage RST_D is low (between t.sub.0 and t.sub.1), capacitor **110-0** is reset to voltage RST_D. The low voltage of RST_D (between t.sub.0 and t.sub.1) is lower than the pinning voltage of photodiode **102-0**. When the voltage RST_D is lower than the pinning voltage, capacitor **110-0** will be reset to RST_D. Resetting capacitor **110-0** to RST_D (while RST_D is lower than the pinning voltage) may be referred to as a hard reset.
(50) At t.sub.1. RST_D is raised to a second, higher level compared to the first, low level between to and t.sub.1. Between t.sub.1 and t.sub.2, capacitor **110-0** is electrically connected to voltage supply **124-2** through transistor **105-0**, photodiode **102-0**, transistor **104-0**, floating diffusion region

118, and reset transistor **106**. However, because RST_D is above the pinning voltage of photodiode **102-0**, the voltage of capacitor **110-0** will increase (e.g., following RST_D) up to but not exceeding the pinning voltage of photodiode **102-0**. In other words, raising RST_D above the pinning voltage between t.sub.1 and t.sub.2 effectively resets capacitor **110-0** to the pinning voltage. Resetting the capacitor **110-0** to the pinning voltage (as between t.sub.1 and t.sub.2) may sometimes be referred to as a soft reset.
(51) Starting at t.sub.2, control signal RST, TX_**0**, and OG_**0** may be brought low (deasserting transistors **106**, **104-0**, and **105-0**, respectively).
(52) The reset operations in FIG. **5** may be referred to as a hard-soft reset (because the capacitor first undergoes a hard reset between t.sub.0 and t.sub.1 and then undergoes a soft reset between t.sub.1 and t.sub.2). Using a hard-soft reset may advantageously remove any residue from the previous frame, which may mitigate noise associated with lag at the capacitor. This example of a reset operation is merely illustrative. It should be understood that other reset operations may be performed if desired (e.g., a soft reset only).
(53) As shown in FIG. **5**, an integration period may follow the reset period. The integration period may occur between t.sub.3 and t.sub.4. Before the start of the integration period, CLG_REF_**0** may optionally be dropped low. CLG_REF_**0** may be kept low throughout the integration period to reduce dark current during the integration period. During the integration time, charge is generated and accumulates (is integrated) in photodiode **102-0**. Transistor **105-0** may be continuously or repeatedly partially pulsed (e.g., by holding OG_**0** at an intermediate voltage) during the integration period to allow charge from photodiode **102-0** to overflow through transistor **105-0** to capacitor **110-0**. As shown in FIG. **5**, transistor **105-0** may optionally be pulsed one or more times during the integration period. Repeatedly pulsing transistor **105-0** (e.g., repeatedly asserting and deasserting the transistor) may ensure effective charge overflow from photodiode **102-0** into capacitor **110-0** while mitigating dark current. Instead of or in addition to pulsing transistor **105-0**, transistor **105-0** may include a buried implant that allows charge from photodiode **102-0** to overflow through transistor **105-0** to capacitor **110-0**.
(54) At t.sub.5, readout operations for the sub-pixel **52-0** are performed. The readout operations may be performed individually for each sub-pixel. In other words, there is a readout period for sub-pixel **52-0** after the integration period, followed by a readout period for sub-pixel **52-1**, followed by a readout period for sub-pixel **52-2**, and followed by a readout period for sub-pixel **52-3**.
(55) The readout operations for sub-pixel **52-0** will now be described. It should be understood that the readout operations for sub-pixel **52-0** use the corresponding control signals for sub-pixels **52-0** (e.g., TX_**0**, OG_**0**, CLG_REF_**0**). The readout operations for the other sub-pixels are the same as described for sub-pixel **52-0**, only using the appropriate control signals (e.g., TX_**1**, OG_**1**, and CLG_REF_**1** for sub-pixel **52-1**).
(56) At t.sub.5, CLG_REF_**0** is raised back high (after optionally being held low through the integration time). At t.sub.6, the RST control signal is pulsed to assert reset transistor **106**. Pulsing the reset transistor resets floating diffusion region **118** to RST_D from voltage supply **124-2**. After resetting the floating diffusion region, the voltage at floating diffusion region **118** is sampled at t.sub.7 (as indicated by the pulse in sample-and-hold SH in FIG. **5**). This sample may be referred to as a reset sample or a photodiode reset sample.
(57) Next, at t.sub.8, the transfer transistor **104-0** is asserted by raising control signal TX_**0**. This transfers charge from photodiode **102-0** to floating diffusion region **118**. After the charge is transferred from the photodiode, the voltage at floating diffusion region **118** is sampled at t.sub.9 (as indicated by the pulse in sample-and-hold SH). This sample may be referred to as a signal sample or a photodiode signal sample.
(58) The readout from photodiode **102-0** therefore includes a signal sample and a reset sample for double sampling. In double sampling, a reset signal and a signal sample are obtained during readout. The reset signal may then be subtracted from the signal sample during subsequent

processing to help correct for noise. The double sampling may be correlated double sampling (in which the reset value is sampled before the signal value) or uncorrelated double sampling (in which the reset value is sampled after the signal value is sampled, sometimes referred to as simply double sampling).

(59) During the readout operations of FIG. **5**, the readout from photodiode **102-0** is a correlated double sampling readout.

(60) After reading photodiode **102-0**, the signal from capacitor **110-0** needs to be read out. Based on the topology of the pixel, the signal from capacitor **110-0** needs to be read out through photodiode **102-0**. To achieve this, at t.sub.10, control signal TX_**0** is raised high to assert transistor **104-0**, control signal OG_**0** is raised high to assert transistor **105-0**, and CLG_REF_**0** is dropped. CLG_REF_**0** may be dropped to an intermediate level, for example. The intermediate level for CLG_REF_**0** may be between the low level used during the integration period and the high level used during the reset period. Asserting transistors **104-0** and **105-0** connects capacitor **110-0** to floating diffusion region **118**. Lowering CLG_REF_**0** at t.sub.10 to the intermediate level may ensure that the signal passes through photodiode **102-0** to floating diffusion region **118**.

(61) After transferring the charge from capacitor **110-0** to floating diffusion region **118**, the voltage at floating diffusion region **118** is sampled at t.sub.11 (as indicated by the pulse in sample-and-hold SH). This sample may be referred to as a signal sample, a capacitor signal sample, or an overflow signal sample.

(62) Next, the floating diffusion region **118** is reset. To mitigate noise, a hard-soft reset procedure is performed (that matches the reset procedure of the reset period between t.sub.0 and t.sub.2). At t.sub.12, RST_D is dropped low while RST, CLF_REF_**0**, OG_**0**, and TG_**0** are kept high. Between t.sub.12 and t.sub.13, capacitor **110-0** is electrically connected to voltage supply **124-2** through transistor **105-0**, photodiode **102-0**, transistor **104-0**, floating diffusion region **118**, and reset transistor **106**. When voltage RST_D is low (between t.sub.12 and t.sub.13), capacitor **110-0** is reset to voltage RST_D. The low voltage of RST_D (between t.sub.12 and t.sub.13) is lower than the pinning voltage of photodiode **102-0**. When the voltage RST_D is lower than the pinning voltage, capacitor **110-0** will be reset to RST_D. The aforementioned operation is the hard reset portion of the hard-soft reset operation.

(63) At t.sub.13, RST_D is raised to a second, higher level compared to the first, low level between t.sub.12 and t.sub.13. Between t.sub.13 and t.sub.14, capacitor **110-0** is electrically connected to voltage supply **124-2** through transistor **105-0**, photodiode **102-0**, transistor **104-0**, floating diffusion region **118**, and reset transistor **106**. However, because RST_D is above the pinning voltage of photodiode **102-0**, the voltage of capacitor **110-0** will increase (e.g., following RST_D) up to but not exceeding the pinning voltage of photodiode **102-0**. In other words, raising RST_D above the pinning voltage between t.sub.13 and t.sub.14 effectively resets capacitor **110-0** to the pinning voltage. The aforementioned operation is the soft reset portion of the hard-soft reset operation.

(64) Starting at t.sub.14, control signal RST, TX_**0**, and OG_**0** may be brought low (deasserting transistors **106**, **104-0**, and **105-0**, respectively). At t.sub.15, control signal TX_**0** is raised high, OG_**0** is raised high, and CLG_REF_**0** is dropped to an intermediate level. The configuration at tis therefore matches the configuration at t.sub.10 (before the signal level is sampled at t.sub.11). At t.sub.16, the voltage at floating diffusion region **118** is sampled. This sample from t.sub.16 may be referred to as a reset sample, a capacitor reset sample, or an overflow reset sample.

(65) During the readout operations of FIG. **5**, the readout from photodiode **102-0** is an uncorrelated double sampling readout (because the reset level is sampled after the signal level).

(66) It is noted that the timing diagram of FIG. **5** may be applied to the pixel of FIG. **3** in addition to the pixel of FIG. **4**. When the timing diagram of FIG. **5** is applied to the pixel of FIG. **3**, the readout of the second, third, and fourth sub-pixels is omitted. The timing diagram depicted in FIG. **5** may be applied to numerous other types of pixels, as will be now described.

(67) One alternate pixel structure is shown in FIG. **6**. The pixel in FIG. **6** is the same as the pixel in FIG. **3**, but with an additional transistor **136** (sometimes referred to as a dual conversion gain transistor) that is be coupled between floating diffusion region **118** and an additional capacitor **138** (sometimes referred to as a dual conversion gain capacitor). The additional capacitor in this arrangement may allow for transferring charge from capacitor **110** to floating diffusion region **118** without charge sharing and without a knee point in the signal range of the floating diffusion region.
(68) Another alternate pixel structure is shown in FIG. **7**. The pixel in FIG. **7** is the same as the pixel in FIG. **4**, but with the additional transistor **136** and capacitor **138** from FIG. **6**. In FIG. **7**, transistor **136** and capacitor **138** are shared between the multiple sub-pixels of the pixel.
(69) Another alternate pixel structure is shown in FIG. **8**. The pixel in FIG. **8** is the same as the pixel in FIG. **3**, but with parallel overflow capacitors. As shown in FIG. **8**, charge from photodiode **102** may first overflow through a first transistor **105-1** to overflow capacitor **110-1**. Capacitor **110-1** is coupled to an associated voltage supply **122-1**. Once capacitor **110-1** is full, charge from photodiode **102** may next overflow through a second transistor **105-2** to overflow capacitor **110-2**. Capacitor **110-2** is coupled to an associated voltage supply **122-2**. During the integration period, there may also be an anti-blooming path to voltage supply **124-2** through transistor **104** and transistor **106**.
(70) Instead of multiple overflow capacitors connected to the photodiode in parallel (as in FIG. **8**), a pixel may have multiple overflow capacitors connected to the photodiode in series, as shown in FIG. **9**. As shown in FIG. **9**, charge from photodiode **102** may first overflow through a first transistor **105-1** to overflow capacitor **110-1**. Capacitor **110-1** is coupled to an associated voltage supply **122-1**. Once capacitor **110-1** is full, charge from photodiode **102** may next overflow through a second transistor **105-2** to overflow capacitor **110-2**. Capacitor **110-2** is coupled to an associated voltage supply **122-2**. During the integration period, there may also be an anti-blooming path to voltage supply **124-2** through transistor **104** and transistor **106**. The rest of the pixel in FIG. **9** is the same as the pixel in FIG. **3**.
(71) In another alternate pixel, shown in FIG. **10**, a coupled gate structure may be included. As shown in FIG. **10**, charge from photodiode **102** may first overflow through a first transistor **105-1** to overflow capacitor **110-1**. Capacitor **110-1** is coupled to an associated voltage supply **122-1**. Once capacitor **110-1** is full, charge from photodiode **102** may next overflow through a second transistor **105-2** to a node **140**. A transistor **105-3** is coupled between node **140** and capacitor **110-2**. A transistor **105-4** is coupled between node **140** and voltage supply **124-2**. Once at node **140**, the overflow charge may either be stored at capacitor **110-2** (when transistor **105-3** is asserted while transistor **105-4** is deasserted) or discarded (when transistor **105-4** is asserted while transistor **105-3** is deasserted). Transistors **105-3** and **105-4** may be asserted in a mutually exclusive fashion. In other words, asserting transistor **105-3** causes transistor **105-4** to be deasserted and asserting transistor **105-4** causes transistor **105-3** to be deasserted. The percentage of time that transistor **105-3** is asserted may be tuned to select the dynamic range of the pixel. The rest of the pixel in FIG. **10** is the same as the pixel in FIG. **3**.
(72) Other arrangements of overflow capacitors in series, overflow capacitor in parallel, and coupled gate structures (where charge is selectively stored or discarded using transistors asserted in mutually exclusively fashion) may be used if desired. In another alternative, a pixel may have a first overflow to a first capacitor and a second, coupled gate overflow from the first capacitor to a second capacitor. In other words, once the first capacitor is full the overflow charge is either discarded or stored in a second capacitor that is in series with the first capacitor. In another alternative, a pixel may have a first, coupled gate overflow to a first capacitor and a second overflow from the first capacitor to a second capacitor. In other words, overflow charge is either discarded or stored in a first capacitor. Once the first capacitor is full, the overflow charge is either discarded or stored in a second capacitor.
(73) The methods of operation described herein may also be used in global shutter pixels, such as

the global shutter pixels described in U.S. patent application Ser. No. 17/811,365, which is hereby incorporated by reference herein in its entirety.
(74) FIG. **11** is a diagram of a pixel with a calibration transistor that may be used during calibration operations for the pixel. As shown in FIG. **11**, the pixel may include a photodiode **102**, transfer transistor **104**, floating diffusion region **118**, source follower transistor **112**, row select transistor **132**, and column output line **116** (similar to as previously discussed in connection with FIG. **3**). However, pixel **34** in FIG. **11** also includes a calibration transistor **142** that is coupled between capacitor **110** and voltage supply **124-2**. Calibration transistor **142** has a gate that receives a control signal CAL.
(75) The pinning voltage of the photodiode in each pixel may vary due to manufacturing variations. Including calibration transistor **142** allows for per-pixel pinning voltage calibration to be performed during operation of the image sensor, which mitigates variability among the pixels. The calibration transistor helps calibrate for a knee point in the response where the floating diffusion regions starts charge sharing with photodiode **102** and capacitor **110**.
(76) FIG. **12** is a timing diagram showing illustrative operation of imaging pixel **34** in FIG. **11**. Between t.sub.0 and t.sub.1, the capacitor **110** and photodiode **102** are both reset by raising control signals OG, TX, and RST, to assert transistors **105**, **104**, and **106**, respectively.
(77) After the reset period there is an integration period between t.sub.1 and t.sub.2. During the integration time, charge is generated and accumulates (integrated) in photodiode **102**. Transistor **105** may be continuously or repeatedly partially pulsed (e.g., by holding OG at an intermediate voltage) during the integration period to allow charge from photodiode **102** to overflow through transistor **105** to capacitor **110**. As shown in FIG. **5**, transistor **105** may optionally be pulsed one or more times during the integration period. Repeatedly pulsing transistor **105** (e.g., repeatedly asserting and deasserting the transistor) may ensure effective charge overflow from photodiode **102** into capacitor **110** while mitigating dark current. Instead of or in addition to pulsing transistor **105**, transistor **105** may include a buried implant that allows charge from photodiode **102** to overflow through transistor **105** to capacitor **110**.
(78) Next, readout operations may be performed. First, a correlated double sampling of the charge in photodiode is performed. At t.sub.2, the reset level of floating diffusion region **118** is sampled (as indicated by the pulse in sample-and-hold-reset SHR). Transfer transistor **104** is then asserted at t.sub.3 to transfer charge from photodiode **102** to floating diffusion region **118**. At t.sub.4, the signal level of floating diffusion region **118** is sampled (as indicated by the pulse in sample-and-hold-signal SHS). These two samples may be used for correlated double sampling of the charge in the photodiode.
(79) Second, a correlated double sampling of the charge in the capacitor is performed. At t.sub.5, reset transistor **106** is asserted. At t.sub.6, the reset level of floating diffusion region **118** is sampled (as indicated by the pulse in sample-and-hold-reset SHR). At t.sub.7, the overflow transistor **105** is asserted (while transistor **104** is asserted) to transfer charge from capacitor **110** to floating diffusion region **118**. The signal level of floating diffusion region **118** is then sampled (as indicated by the pulse in sample-and-hold-signal SHS). These two samples may be used for correlated double sampling of the charge in the capacitor.
(80) Finally, a calibration operation may be performed between t.sub.8 and t.sub.11. At t.sub.8, RST_D is dropped to a voltage that is below the pinning voltage of photodiode **102**. At the same time, calibration transistor **142** and reset transistor **106** are asserted (by raising control signals CAL and RST respectively). Next, at t.sub.9, transistor **105** is asserted (while transistor **104** is asserted). Finally, at t.sub.10, the signal level of floating diffusion region **118** is sampled (as indicated by the pulse in sample-and-hold-signal SHS). This calibration operation allows for the pinning voltage of the photodiode to be read out from the floating diffusion region.
(81) If desired, any of the pixels herein may be split between two or more stacked substrates. In FIG. **3**, a conductive interconnect may be positioned between transistor **104** and floating diffusion

region **118** at location **134** (with the components to the left of location **134** in FIG. **3** incorporated in a first semiconductor substrate and the components to the right of location **134** in FIG. **3** incorporated in a second semiconductor substrate). In FIG. **4**, a conductive interconnect may be positioned at location **134** between transistors **104-0**, **104-1**, **104-2**, and **104-3**, and floating diffusion region **118**. Floating diffusion region **118**, reset transistor **106**, source follower transistor **112**, row select transistor **132**, column output line **116**, and voltage supplies **124-2** and **124-3** may be incorporated in a first semiconductor substrate whereas the remaining components of the four respective sub-pixels may be incorporated in a second semiconductor substrate.
(82) The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. A method of operating an imaging pixel that comprises a photodiode, a floating diffusion region, a first transistor coupled between the photodiode and the floating diffusion region, a capacitor, a second transistor coupled between the photodiode and the capacitor, a voltage supply, and a third transistor coupled between the floating diffusion region and the voltage supply, the method comprising: during a first portion of a reset period, asserting the first, second, and third transistors while the voltage supply provides a voltage at a first magnitude that is less than a pinning voltage of the photodiode; and during a second portion of a reset period, asserting the first and third transistors while the voltage supply provides the voltage at a second magnitude that is greater than the pinning voltage of the photodiode.
2. The method defined in claim 1, further comprising: during an integration period, storing charge at the photodiode and the capacitor.
3. The method defined in claim 2, further comprising: during the integration period, pulsing the second transistor to an intermediate level.
4. The method defined in claim 3, further comprising: during a readout period that is subsequent to the integration period, sampling a first reset level.
5. The method defined in claim 4, further comprising, during the readout period: asserting the first transistor to transfer charge from the photodiode to the floating diffusion region; and sampling a first signal level.
6. The method defined in claim 5, further comprising, during the readout period and after asserting the first transistor to transfer charge from the photodiode to the floating diffusion region: asserting the first transistor and the second transistor; and sampling a second signal level.
7. The method defined in claim 6, further comprising, during the readout period and after sampling the second signal level: asserting the first, second, and third transistors while the voltage supply provides the voltage at the first magnitude; asserting the first, second, and third transistors while the voltage supply provides the voltage at the second magnitude; and sampling a second reset level.
8. The method defined in claim 1, wherein the imaging pixel comprises an additional voltage supply and wherein the capacitor is coupled between the additional voltage supply and the second transistor.
9. The method defined in claim 8, further comprising: during the reset period and using the additional voltage supply, providing an additional voltage at a third magnitude.
10. The method defined in claim 9, further comprising: during the integration period and using the additional voltage supply, providing the additional voltage at a fourth magnitude that is less than the third magnitude.
11. The method defined in claim 10, further comprising: during a readout period and using the additional voltage supply, providing the additional voltage at a fifth magnitude that is less than the third magnitude.

12. The method defined in claim 11, wherein the fifth magnitude is greater than the fourth magnitude.
13. The method defined in claim 1, further comprising: asserting the second transistor during the second portion of the reset period.
14. An imaging pixel, comprising: a photodiode that has a pinning voltage; a floating diffusion region; a first transistor coupled between the photodiode and the floating diffusion region; a capacitor; a second transistor coupled between the photodiode and the capacitor; a voltage supply configured to provide a voltage; and a third transistor coupled between the floating diffusion region and the voltage supply, wherein the voltage supply is configured to, during a first portion of a reset period, provide the voltage at a first magnitude that is less than the pinning voltage and wherein the voltage supply is configured to, during a second portion of the reset period, provide the voltage at a second magnitude that is greater than the pinning voltage.
15. The imaging pixel defined in claim 14, further comprising: an additional photodiode that has an additional pinning voltage; a fourth transistor coupled between the additional photodiode and the floating diffusion region; an additional capacitor; and a fifth transistor coupled between the additional photodiode and the additional capacitor, wherein the first magnitude is less than the additional pinning voltage and wherein the second magnitude is greater than the additional pinning voltage.
16. The imaging pixel defined in claim 14, further comprising: an additional capacitor; a fourth transistor coupled between the additional capacitor and the capacitor, wherein the additional capacitor is connected to the photodiode in series with the capacitor.
17. The imaging pixel defined in claim 14, further comprising: an additional capacitor; a fourth transistor coupled between the additional capacitor and the photodiode, wherein the additional capacitor is connected to the photodiode in parallel with the capacitor.
18. The imaging pixel defined in claim 14, further comprising: an additional capacitor; a fourth transistor coupled between the additional capacitor and the floating diffusion region.
19. The imaging pixel defined in claim 14, further comprising: a first semiconductor substrate that includes the photodiode, the first transistor, the capacitor, and the second transistor; a second semiconductor substrate that includes the floating diffusion region, the voltage supply, and the third transistor; and a conductive interconnect that electrically connects the first transistor in the first semiconductor substrate to the floating diffusion region in the second semiconductor substrate.
20. A method of operating an imaging pixel that comprises a photodiode, a floating diffusion region, a first transistor coupled between the photodiode and the floating diffusion region, a capacitor, a second transistor coupled between the photodiode and the capacitor, and a voltage supply, wherein the capacitor is coupled between the voltage supply and the second transistor, wherein the first and second transistors are connected in series between the capacitor and the floating diffusion region, and wherein the method comprises: during a reset period and using the voltage supply, providing a voltage at a first magnitude; during an integration period, storing charge at the photodiode and at the capacitor; and during a readout period and using the voltage supply, providing the voltage at a second magnitude that is less than the first magnitude.
21. The method defined in claim 20, wherein the photodiode is directly connected to a node and wherein the node is interposed between the first and second transistors.
22. The method defined in claim 20, wherein the capacitor has a first plate coupled directly to the voltage supply and wherein the capacitor has a second plate coupled directly to the second transistor.